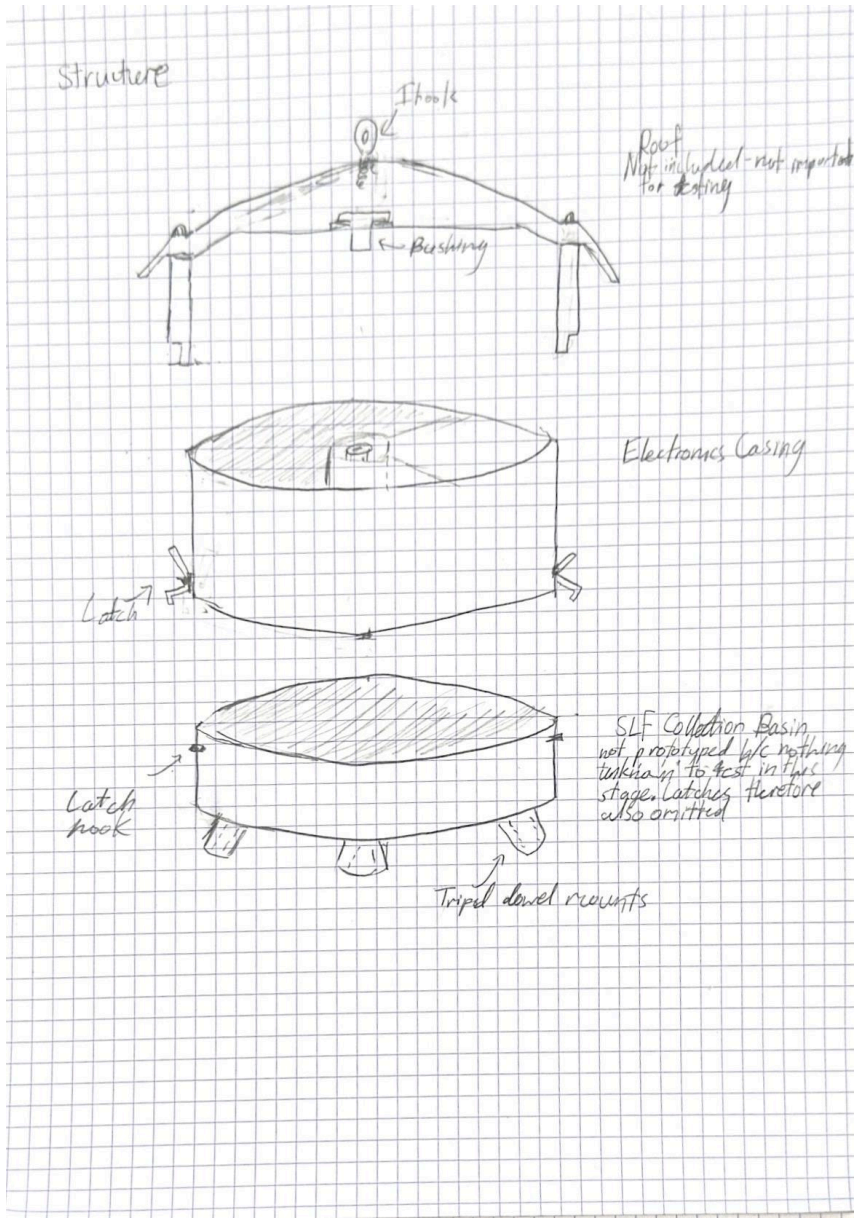
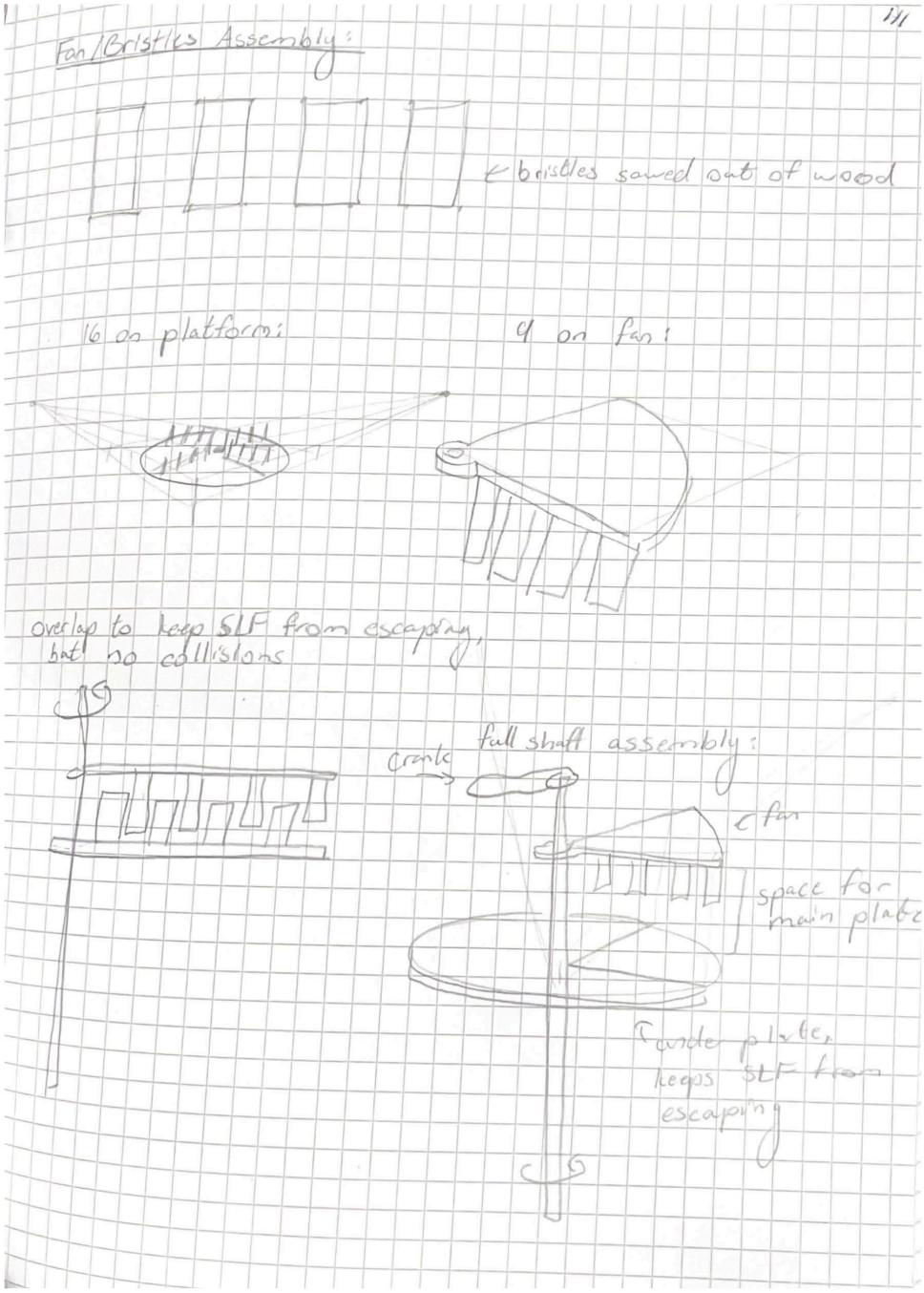
Design Documentation

Part	Specs	Fabrication Details
Sweeping Rotor	⅛ inch thick balsa wood, quarter of a 10cm radius circle.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Teeth cut out of wood using a bandsaw/hand saw and hot glued/gorilla glued onto the disk. Hole drilled in the middle for the dowel.
Spinning Open Disk	⅛ inch thick balsa wood, 21cm in diameter.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Hole drilled in the middle for the dowel.
SLF Platform	⅛ inch thick balsa wood, 22cm in diameter.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Teeth cut out of wood using a bandsaw/hand saw and hot glued/gorilla glued onto the disk
Rod	⅜ inch dowel	Cut to size
Crank	⅛ inch thick balsa wood, in an oval shape	Cut out of wood using bandsaw and sanded down using sander. Hole drilled in the middle for the dowel. Connected the crank to the dowel with hot glue and zip ties.
Electronics Casing	23x23x10cm, PLA	3D Printed by RPL

Sketches:

Structure


Hook
Roof
Not included - not important for testing
Bushing
Electronics Casing
Latch
SLF Collection Basin
not prototyped b/c nothing tuskin' to test in this stage. Latches therefore also omitted
Latch hook
Tripod dowel mounts



Assembly:

Part 1: Cut the Main Wooden Pieces

- Step 1: Draw and cut the base shapes
  - From the balsa wood, draw and cut:
    - one 22 cm diameter circle for the SLF Platform
    - one 21 cm diameter circle for the Spinning Open Disk
    - one 7 cm × 3 cm oval for the crank

        - one quarter circle with 10 cm radius for the sweeping rotor
        - fifteen 1 cm × 4 cm rectangles for the teet
    - Try to keep all cuts as clean and square as possible so the parts align well during assembly.

Part 2: Prepare the SLF Platform

- Step 2: Divide the platform into four equal sections
    - Take the 22 cm SLF Platform disk.
    - Draw one straight line through the center of the circle.
    - Draw a second straight line through the center, perpendicular to the first.
    - This divides the disk into 4 equal quadrants
    - These center lines will be used to place the teeth slots.
- Step 3: Mark the tooth slots
    - Along each of the four center lines:
    - Mark 3 evenly spaced rectangular slots.
    - Each slot should measure approximately:
        i. 1 cm long
        ii. 1/8 inch wide
    - The center line should form the base/alignment line for the slots.
    - This creates a total of 12 slots on the SLF Platform.
    - Space them evenly from the center outward so they are distributed consistently across the disk.
- Step 4: Mark the open areas
    - On the opposite side of each line, mark larger rectangular openings extending away from the center line.
    - leave about 1 cm of material between the opening and:
        - the center line
        - the outer edge of the disk
    - These cutouts create open areas that allow swept insects to move downward into the trap.
- Step 5: Cut the marked openings
    - Carefully cut out:
        - the 12 small tooth slots
        - the larger rectangular openings
    - Clean all edges so the slots are wide enough for the teeth pieces to fit snugly.
- Step 6: Install the platform teeth
    - Insert the rectangular teeth into the small slots on the SLF Platform.
    - Each tooth should stand vertically.
    - Apply wood glue at the base of each tooth.
    - Make sure all teeth are aligned in the same direction and stand upright while drying.

Part 3: Prepare the Sweeping Rotor

- Step 7: Transfer tooth positions to the rotor
    - Take the quarter-circle sweeping rotor.
    - To align its teeth with the platform:

        - Place the sweeping rotor directly on top of the completed SLF Platform.
        - Position it so the curved outer edge of the rotor lines up with the row of platform teeth.
        - Use the locations of the platform teeth as a guide to mark corresponding tooth positions on the rotor.
    - Important: The rotor teeth should be placed in the opposite / complementary positions to the platform teeth so that when the rotor turns, it sweeps between or against the platform features without major interference.
- Step 8: Cut rotor tooth slots
    - At the marked locations on the sweeping rotor:
        - cut rectangular slots sized to fit the teeth pieces
        - use the same slot size as before:
            - about 1 cm long
            - 1/8 inch wide
- Step 9: Install rotor teeth
    - Insert the remaining teeth pieces into the sweeping rotor slots.
    - Glue each tooth upright with wood glue.
    - Keep spacing and alignment as even as possible.
    - Let the glue dry fully before assembly

Part 4: Prepare the Spinning Open Disk

- Step 10: Cut out the open section
    - Take the 21 cm Spinning Open Disk.
        - Mark a quarter-circle section to remove.
        - Cut out that quarter section.
        - Do not remove the center area around the shaft hole, leave enough material near the center so the disk remains structurally connected and can rotate on the dowel.
        - This creates an open region in the lower rotating disk while maintaining support around the shaft.

Part 5: Drill Center Holes

- Step 11: Drill shaft holes
    - Using a 3/8 inch drill bit, drill a centered hole in each of the following:
        - Sweeping Rotor
        - SLF Platform
        - Spinning Open Disk
        - Crank
    - Make sure:
        - each hole is centered as accurately as possible
        - the dowel can pass through each hole
        - the fit is not so tight that rotation is restricted

Part 6: Print the Electronics Casing

- Step 12: 3D print the casing
    - 3D print the cylindrical electronics casing in PLA.

    - This printed part serves as the lower housing/base and supports the rotating assembly above it.
    - The finished printed casing should match the cylindrical form shown in the photos.

Part 7: Final Assembly

- Step 13: Insert the dowel into the casing
    - Place the wooden dowel vertically through the center of the 3D-printed electronics casing,
    - Make sure the dowel is upright and centered.
- Step 14: Attach the Spinning Open Disk
    - Slide the Spinning Open Disk onto the dowel so it sits at the top of the electronics casing.
    - Secure it in place by:
        - placing one zip tie below the disk
        - placing one zip tie above the disk
        - applying wood glue where needed for extra holding strength
    - This disk should be supported on the dowel and positioned just above the casing.
- Step 15: Position the SLF Platform
    - Slide the SLF Platform onto the dowel above the Spinning Open Disk
    - Position it so that:
        - it sits securely over the casing area
        - it is stable
        - the dowel can still rotate freely relative to the platform as intended by your design
- Step 16: Attach the Sweeping Rotor
    - Slide the Sweeping Rotor onto the dowel above the SLF Platform.
    - Position it so that:
        - it sits just above the platform
        - its teeth can sweep across or near the platform teeth/openings
        - there is slight clearance to avoid jamming
    - Secure the rotor with:
        - one zip tie below it
        - one zip tie above it
        - wood glue for reinforcement
- Step 17: Attach the crank
    - Slide the crank onto the dowel near the top.
    - place the crank about 1 cm above the sweeping rotor, or
    - at a convenient height near the top of the dowel for manual turning
    - Secure the crank with:
        - one zip tie below
        - one zip tie above
        - wood glue

Design Test

Question 1: Will the fan move smoothly through the bristles?

We compared the force it takes to pull the fan through the flat regions and the bristled parts. The hook for the force meter was always right next to the outermost bristle of the fan. The force meter maxes out at 250 g.

Flat region	Force required take 1 (g)	Force required take 2 (g)	Force required take 3 (g)	Force required avg (g)
1	76	105	98	93
2	105	99	96	100
3	139	129	119	129
4	123	60	88	90
Total avg	-	-	-	103

teeth	Force required take 1 (g)	Force required take 2 (g)	Force required take 3 (g)	Force required avg (g)
1	89	96	88	91
2	138	200	136	158
3	158	151	175	161
4	>250	>250	>250	>250
Total avg	-	-	-	165

The average force required to move the fan through the bristles is 60% greater than the average force to move the fan elsewhere, so the fan is not moving smoothly through the bristles. Major factors contributing to this is the imprecision in bristle sizes due to handsawing, the top of the fan interacting with the glue at the bottom of the bristles, and the bending in the fan (further discussed later).

For the final prototype, we will definitely not use wood and instead opt for a more flexible material. We plan on testing TPU 3D print filament and, if the TPU isn't precise enough, using store-bought plastic combs. In the event that we do use filament, the shape of the bristles will also be more curved to account for the curvature of the fan's path.

Question 2: Will we able to reduce the speed of the fan to 1-2 rpm?

Due to motor expenses and limited fabrication time, we decided to hold off on installing a motor for this prototype, and we will test this at a later date. Hand cranking allowed a maximum speed of around 10 rpm and a minimum speed of around 1 rpm.

Question 3: Will there be little friction between the fan and the plate?
On the flat region of the upper plate, the bending of the outer parts of the fan caused the fan to rub against the plate. This was especially noticeable in regions 2 and 3, which is why the forces to pull them across these areas were greater than the other two. The bottom plate and the upper plate are separated by a ziptie, and we observed some slight resistance between the two plates indicating friction. The frictional coefficient of the balsa wood that we were using is between 0.25-0.5. The frictional coefficient of other materials we are considering include:

Balsa Wood: 0.24-0.5
PLA: 0.52-0.67 for 3D printed PLA
ABS: 0.4-0.5 for 3D printed ABS
PETG: 0.25-0.40 for 3D printed PETG

Overall, the frictional coefficient for balsa wood is on the lower end for our potential materials which means it remains a leading choice for our final design. We should also focus on ensuring the plates are designed in a way such that they do not touch or have any friction in the final design. The bending of the fan still needs to be addressed (see next question).

Frictional coefficient data:
https://www.sciencedirect.com/science/article/pii/S175161612100182X -balsa
https://iopscience.iop.org/article/10.1088/1742-6596/2542/1/012003/pdf - pla
https://pmc.ncbi.nlm.nih.gov/articles/PMC12298762/ -abs
https://tribology.rs/journals/2024/2024-1/9-1546.pdf- petg

Question 4: Will weight cause the outer parts of the fan and bottom plate to bend?
The vertical displacement between the center and end of the mechanical fan was originally 16 mm. After gluing down the fan, there was still a vertical displacement of 7mm.
The bottom plate was resting on the motor casing, so it was not bending. However, the wood did seem to be warped, as one side was higher than the other when it was only on the shaft by around 2 mm.
For the final prototype, we will definitely want a bearing to help hold the fan up better. We can also let the edge of the fan rest on the lip of the prototype's casing if we determine that doing so will not cause enough friction to inhibit the motor.

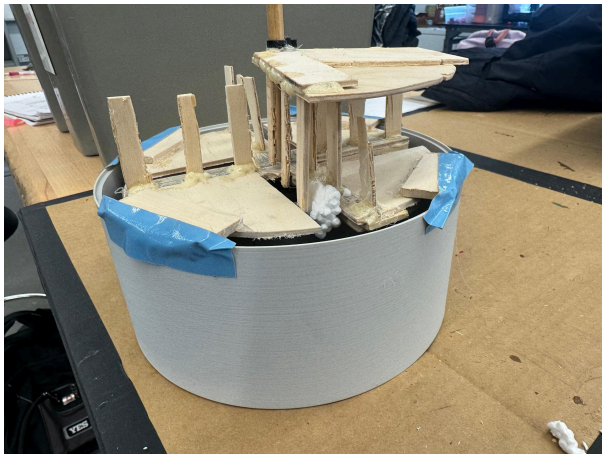
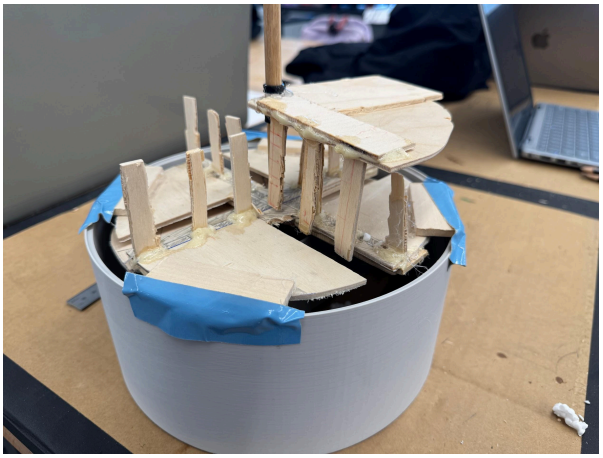
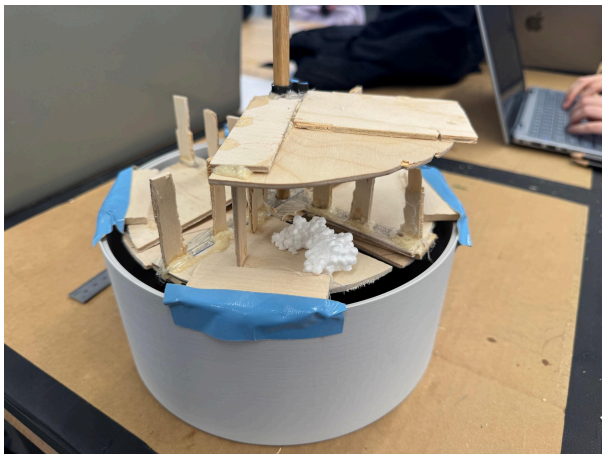
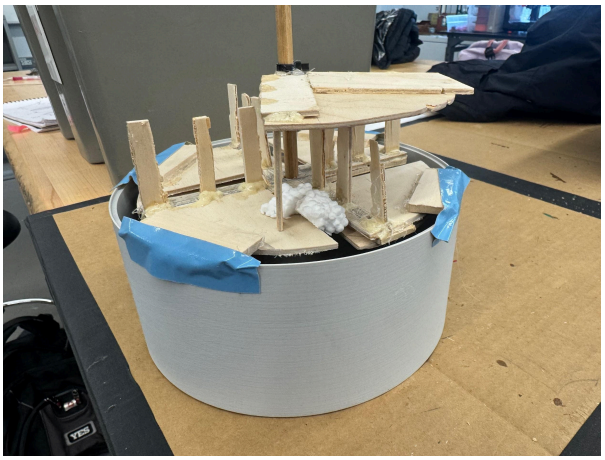
Question 5: How does the locking mechanism for the basin feel? Is it easy to remove?
We decided that since this locking would just be buying and bolting in industry standard latches from McMaster, it wouldn't be worth our time to test if they're "good" enough. We decided instead to spend our time testing our mechanism.

Bonus: ability to move SLF-sized objects
We modelled the SLF as shredded pieces of foam, probably about 1g each. The teeth were strong enough to move the foam pieces into the slots. At the moment, though, the teeth are

each about 1.5cm apart, which is big enough to miss a full-grown SLF entirely. This also goes back to the materials we choose to make the fan teeth out of, as discussed in the first question.

To test, we ran three different trial runs with 10 pieces of foam about the size of an SLF. We found that there was a percent error of 36.7% due to the way the slots were made. To fix this issue, we will slightly increase the gap size of the trap for the SLF to fit into. We expect non-trapped SLF to simply be crushed due to the high gear ratio of the motor, but still wish to limit this since we're worried about clogging over time.

Total "SLF" Tested	**"SLF" that were caught**
10	7
10	6
10	6
Average: 10	**Average:** 6.33

<u>Success Criteria</u>

1. The fan can carry objects of mass up to 5g and diameter as small as 5mm (approximately SLF-sized objects) into the slots.
2. Fan with bristle spacing small enough to meet above requirements can move through static bristles on the SLF platform with maximum of 2.356 N*m of torque
3. The motion of the fan reliably continues for at least 30 consecutive revolutions without observed scratching or deformity to ensure that its operation is repeatable over time.